

Self-Interacting Stochastic Dynamics with Exponential Memory: An Observable-Matched Linear Relaxation Baseline

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We study a stochastic trajectory coupled to a spatial trace with exponential forgetting. A compact trace moving with the trajectory can resemble self-localization even when its size is controlled by linear feedback. We distinguish the current position relative to the trace centroid from the width of the trace itself and derive their separate stationary second moments. For a finite history, an impulse-response calculation and an independent spectral integral give matching linear references without fitted parameters. Reanalysis of nine active long-run slices, with five seeds each, dimensions $d = 3\text{--}20$, and runs up to 3×10^8 updates, gives a median memory-cloud radius discrepancy of 0.076% and a maximum of 0.984% in the stored terminal windows. Curvature-matched one- and two-scale Gaussian kernels yield nearly identical scalar observables in the small-radius regime. A separate fixed-gain test gives a smooth 6.24% radius correction as the sampled scale increases, without resolving a metastable transition. These results establish an observable-specific baseline for the tested branch and delimit what compact memory clouds can identify about the interaction kernel.

I. INTRODUCTION

Motion influenced by its own history occurs in reinforced walks, Brownian polymers, and self-interacting diffusions [1, 2]. For normalized cumulative occupation measures, Benaïm, Ledoux, and Raimond developed a diffusion framework; symmetric interactions can drive the occupation measure toward the critical set of a non-linear free-energy functional [3, 4]. Exponentially fading traces replace cumulative averaging by a fixed persistence scale. Aging-weighted diffusions with linear history interactions, including an exponential case, have been studied by Milišić, Meunier, and Roux [5].

The physical architecture of self-deposition and subsequent field response also has established precedents. Walking-droplet models couple a moving source to its decaying wave field [6]; autochemotaxis couples stochastic motion to a chemical signal and distinguishes transient self-localization from long-time transport [7]. Those models include physical field propagation or diffusion and, in the droplet case, inertial motion. Here the dynamics is a discrete overdamped map with a nonpropagating deposited trace and a prescribed interaction kernel. Adding auxiliary state to represent memory is likewise an established method [8, 9].

The contribution is a quantitative diagnostic for this specific map. We match the theoretical and measured radius observables, including finite history and averaging conventions, then test the same fixed reference against archived long runs and kernel controls. This distinction matters: the instantaneous width of a trace is not the distance of its most recent point from its centroid, and a median instantaneous width is not a stationary root-mean-square (RMS) radius. Agreement between different quantities can obscure model and measurement errors.

II. DEPOSITED-TRACE MODEL

On $\mathbb{R}^d$, let μ_n be a finite nonnegative memory measure and K a smooth translation-invariant interaction potential. We require finite second moments and integrable K and ∇K against the memory. We use

$$x_{n+1} = x_n - \eta \nabla \Phi_n(x_n) + \varepsilon \xi_n, \quad (1)$$

$$\Phi_n(x) = \int K(x - y) \mu_n(dy), \quad (2)$$

$$\mu_{n+1} = q\mu_n + \alpha \delta_{x_{n+1}}, \quad q = 1 - \alpha, \quad (3)$$

where $0 < \alpha < 1$, $\eta \geq 0$, and the independent innovations satisfy $\xi_n \sim \mathcal{N}(0, I_d)$. The gradient acts on the evaluation coordinate x in Eq. (2); integrating over y does not remove this dependence. A unit-mass initial memory preserves unit mass under Eq. (3). A centered deposition profile can replace the point measure, but every simulation reported here uses point deposition.

The untruncated augmented state (x_n, μ_n) is Markov: it determines the next-state law given a fresh innovation. The visible coordinate generally fails this property because the same position with a different memory can have a different drift. Exponential discounting supplies a persistence scale α^{-1} updates. It need not destroy information in an exact field representation, and we make no thermodynamic irreversibility claim.

The simulations retain the latest H positions, with

$$\mu_n^H = \alpha \sum_{j=0}^{H-1} q^j \delta_{x_{n-j}}, \quad M_H = 1 - q^H. \quad (4)$$

After the buffer fills, its mass is M_H , not one. The complete finite Markov state is the ordered history buffer. The native update evaluates the force first, advances x , deposits x_{n+1} , and retires the oldest position. This convention fixes the timing in the comparisons below.

The kernel used in the long runs is

$$K(u) = A_r e^{-|u|^2/(2s_r^2)} - A_a e^{-|u|^2/(2s_a^2)}. \quad (5)$$

Its local restoring curvature is $\kappa = A_a/s_a^2 - A_r/s_r^2$. In a region where the pairwise separations are small compared with the active kernel widths, the drift becomes

$$-\eta \nabla \Phi_n(x_n) \simeq -g(x_n - c_n), \quad g = \eta M \kappa, \quad (6)$$

with c_n the mass-normalized memory centroid and M its mass. This local approximation does not imply global confinement.

III. TWO RADIUS OBSERVABLES

For a normalized memory, define

$$r_n = x_n - c_n, \quad Q_n = \int |y - c_n|^2 \mu_n(dy). \quad (7)$$

The position-relative RMS is $R_r = \sqrt{\mathbb{E}|r_n|^2}$; the memory-cloud RMS is $R_Q = \sqrt{\mathbb{E}Q_n}$. Their stationary expectations concern the relative state, not a stationary absolute position.

A. Untruncated reference

For Eq. (3), the centroid obeys $c_{n+1} = qc_n + \alpha x_{n+1}$. Substituting the linear force gives

$$r_{n+1} = q(1 - g)r_n + q\varepsilon \xi_n. \quad (8)$$

If $|q(1 - g)| < 1$, the stationary position-relative second moment is

$$R_r^2 = \frac{dq^2 \varepsilon^2}{1 - q^2(1 - g)^2}. \quad (9)$$

The variance of the deposited trace satisfies a different recursion:

$$Q_{n+1} = qQ_n + q\alpha|x_{n+1} - c_n|^2. \quad (10)$$

Since $r_{n+1} = q(x_{n+1} - c_n)$, stationarity gives

$$R_Q^2 = R_r^2/q. \quad (11)$$

The factor is small at $\alpha = 0.01$, but the distinction remains exact. Moreover, neither RMS equals the median of $\sqrt{Q_n}$.

B. Finite-history reference

The normalized weights are $w_j = \alpha q^j / M_H$. The exact finite centroid update is

$$c_{n+1}^H = qc_n^H + \frac{\alpha}{M_H} (x_{n+1} - q^H x_{n-H+1}). \quad (12)$$

Replacing the mass in g does not remove the retirement term. For the linear map, define its normalized history filter and response denominator on the unit circle by

$$B_H(z) = \sum_{j=0}^{H-1} w_j z^{-j}, \quad (13)$$

$$F_H(z) = 1 - z^{-1} + gz^{-1}[1 - B_H(z)]. \quad (14)$$

In the stationary relative regime the two matching second moments are

$$R_{r,H}^2 = \frac{d\varepsilon^2}{2\pi} \int_{-\pi}^{\pi} \frac{|1 - B_H(e^{i\omega})|^2}{|F_H(e^{i\omega})|^2} d\omega, \quad (15)$$

$$R_{Q,H}^2 = \frac{d\varepsilon^2}{2\pi} \int_{-\pi}^{\pi} \frac{1 - |B_H(e^{i\omega})|^2}{|F_H(e^{i\omega})|^2} d\omega. \quad (16)$$

The apparent zero-frequency singularities are removable for these relative observables. The numerator in Eq. (16) expresses the weighted variance across historical positions, rather than a current-point displacement. For the tested gains $0 \leq g < 1$, the relative state is stable; the unpinned translation mode is retained. For example, in the linear model the per-coordinate long-time diffusion coefficient is

$$D_{\text{eff},H} = \frac{\varepsilon^2}{2(1 + g\bar{j}_H)^2}, \quad \bar{j}_H = \sum_j j w_j. \quad (17)$$

Compactness about the moving centroid is therefore compatible with unbounded large-scale motion. Equation (17) is an analytical linear-model result; we do not fit or validate it with the radius data.

We evaluate Eqs. (15) and (16) independently by a spectral integral and the squared causal impulse response. Doubling the latter from 20,000 to 40,000 updates changes the second moments by less than 3×10^{-12} relatively in the three tested gain cells. The independent calculations agree within 6×10^{-12} . These numerical checks are not interval-certified remainder bounds. At $\alpha = 0.01$, $H = 600$, the finite-history change in the mass-corrected position-relative RMS is at most 0.0074% for the active slices. The exact observable match, rather than this small correction, is the decisive change to the radius comparison.

IV. NUMERICAL EVIDENCE

A. Matched terminal-window comparison

We reanalyze nine archived active slices and eight no-feedback controls. Each slice has seeds 1–5. All use $\alpha = 0.01$, $H = 600$, $\varepsilon = 10^{-4}$, $s_r = 1$, $s_a = 3$, $A_r = 1$, and nominal memory mass one; the active gain uses $\eta = 0.15$, while controls use $\eta = 0$. Table I lists dimensions, attraction amplitudes, and run lengths. The controls shared by the two $d = 3$ attraction slices are counted once.

TABLE I. Median measured/predicted RMS ratios for the archived terminal windows. IDs 0–8 have feedback; IDs 9–16 are no-feedback controls. Common parameters are specified in the text; “off” denotes $\eta = 0$.

ID	d	$N/10^6$	A_a	$R_Q/R_{Q,H}$	$R_r/R_{r,H}$
0	3	30	20	1.0098	1.0076
1	3	30	35	1.0045	1.0037
2	4	30	35	1.0018	1.0019
3	5	30	35	1.0013	1.0012
4	7	30	35	1.0001	1.0001
5	10	30	35	1.0000	1.0001
6	13	30	35	0.9997	0.9998
7	20	30	35	0.9992	0.9991
8	10	300	35	1.0001	1.0000
9	3	30	off	1.0094	1.0087
10	4	30	off	0.9926	0.9872
11	5	30	off	1.0092	1.0079
12	7	30	off	0.9922	0.9914
13	10	30	off	0.9937	0.9956
14	13	30	off	0.9983	0.9988
15	20	30	off	1.0027	1.0019
16	10	300	off	1.0212	1.0198

The stored trace contains 10,001 consecutive terminal measurements spanning 10,000 updates, supplemented by sparse earlier snapshots. We select the maximal uniformly sampled terminal block and discard the sparse points. For each seed s , we compute

$$\hat{R}_{Q,s} = \sqrt{\frac{1}{T} \sum_{n \in \mathcal{W}_s} Q_n}, \quad \hat{R}_{r,s} = \sqrt{\frac{1}{T} \sum_{n \in \mathcal{W}_s} |r_n|^2}, \quad (18)$$

where $T = 10,001$ and $\mathcal{W}_s$ is the terminal block. This uses the same second moments as the theory. We then report the median ratio over five seeds, with the seed interquartile range (IQR). The IQR is dispersion, not a confidence interval; neighboring measurements are correlated. The window spans 100 memory-persistence times even when the preceding run is much longer. These are post-hoc analyses of existing runs, not a new prospective holdout or an equivalence test.

The active memory-cloud ratios in Fig. 1(a) have median absolute departure 0.076% and maximum 0.984%; the corresponding position-relative departures are 0.090% and 0.761%. The maximum cloud departure in the no-feedback controls is 2.122%. The controls are evaluated with their own $g = 0$ reference, rather than an active-force formula. All model coefficients come from the archived configurations. No gain, radius, or covariance coefficient is fitted to these terminal measurements. The comparison supports consistency with a linear relaxation cloud at these parameters. It does not prove global confinement, a complete phase diagram, or exact equality of the nonlinear and linear processes.

B. Kernel and scale controls

The first control compares attractive-only kernels with two-scale kernels at equal local curvature, using $d = 3$, $N = 300,000$, $\varepsilon = 10^{-4}$, and seeds 1–5. With $s_a = 3$ and $s_r = 1$, the effective amplitude of the two-scale family is $A_{\text{eff}} = A_a - 9A_r$. The matched attractive-only family has $A_a = A_{\text{eff}}$, $A_r = 0$. For nine sampled positive effective amplitudes, the 45 seed-paired cloud radius comparisons differ by at most 3.37×10^{-6} relatively. Figure 2(a) shows the paired curves and seed IQRs. These legacy control radii are medians of instantaneous cloud widths in sparse traces. Both families use the same statistic, so their comparison is valid, but these medians are not used to validate Eqs. (16) and (18). In this sampled small-radius regime, compactness does not identify the repulsive component separately from the net curvature.

The second control holds the attractive-only gain fixed at $g = 0.432291\dots$ and increases the nominal position-relative R_r/s_a from 0.03 through 0.1 to 0.3 by changing ε . The three amplitudes are approximately 0.04341, 0.14471, and 0.43412; $d = 3$, $N = 300,000$, $A_a = 26$, and $A_r = 0$ are fixed. The seed-paired endpoint departure in Fig. 2(b) is 6.24%. Normalization to each seed’s low-scale value cancels a constant linear conversion between cloud and position radii. The shape observables show no coordinated transition at the sampled points. A subsequent measurement audit found that fixed-voxel residence changed with cloud scale, while a co-moving residence statistic saturated in both active and no-feedback cases. The original preregistered composite classification therefore remains *inconclusive*; the post-hoc audit does not replace it. Three sampled scales cannot exclude an unsampled bifurcation or a different nonlinear branch.

V. DISCUSSION

The simplest explanation consistent with the tested scalar branch is a linear relative relaxation mode with a freely translating centroid. The deposited-trace width and current-point displacement have different stationary second moments, and comparing the correct quantities restores agreement for both feedback and no-feedback cases. The matched-kernel control further shows that the local curvature, rather than the detailed two-scale structure, determines the measured compactness in the sampled Taylor regime. The fixed-gain control identifies a smooth finite-scale departure without isolating a metastable transition.

This evidence is limited to the archived parameters and terminal windows. It does not establish that every realization is trapped, that the entire nonlinear kernel has a stationary relative distribution, or that nonlinear relative equilibria are absent elsewhere. Finite-history rotating states are a separate deterministic question and are not needed for the present claim. Likewise, a covariance participation dimension near d is an expected isotropic

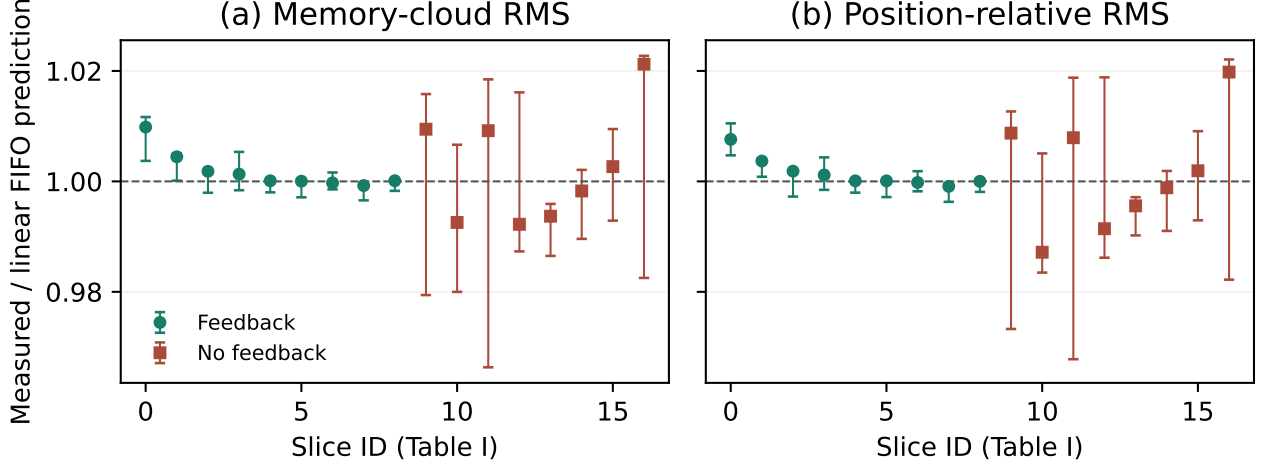


FIG. 1. Observable-matched radius comparison with the linear finite-history reference. (a) Memory-cloud RMS from the weighted trace covariance. (b) Current-position RMS relative to the trace centroid. Symbols are medians over five seeds, bars their IQR, and the dashed line denotes the fixed prediction. Slice IDs and parameters appear in Table I. All measurements use 10,001 terminal snapshots per seed; the earlier run length is not the number of independent measurements.

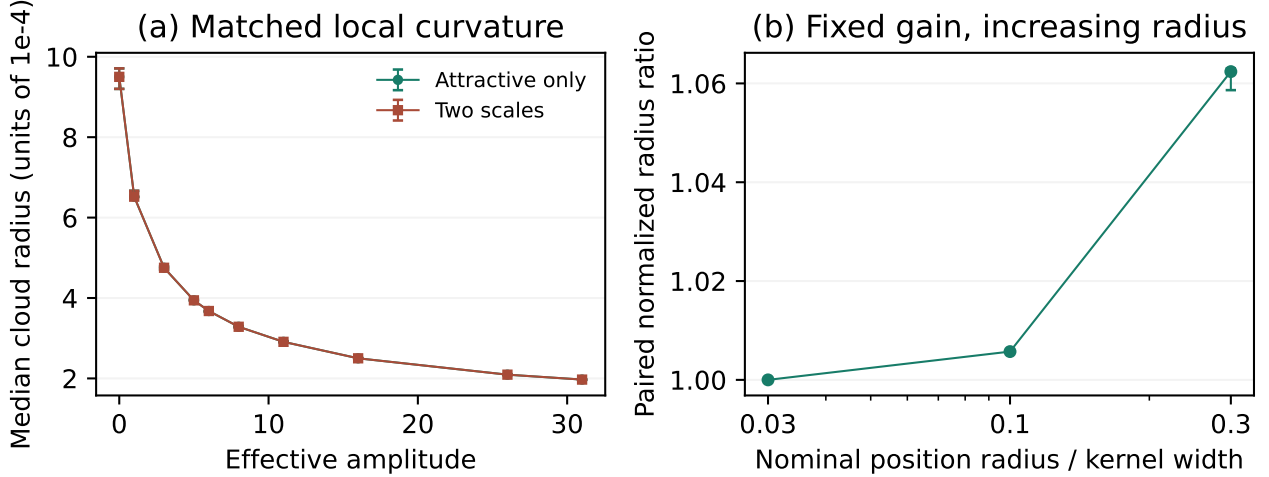


FIG. 2. Independent limits on the interpretation of compact clouds. (a) Attractive-only and two-scale kernels matched by local curvature; points are seed medians of the same legacy cloud-width statistic, with seed IQRs. Their overlap tests local kernel identifiability. (b) At fixed gain, increasing the nominal position-relative radius relative to the kernel width yields a smooth correction. Each seed's normalized cloud median is divided by its low-scale value before aggregation; the endpoint median ratio is 1.0624. Bars are seed IQRs. This panel tests a departure from scale invariance, not an absolute RMS prediction.

shape diagnostic in a supplied d -dimensional embedding; it does not establish dimension selection.

The useful consequence is a testable baseline: evidence for a stronger nonlinear state must depart from the observable-matched linear prediction and survive controls with matched length and time scales. The present results concern stochastic dynamics and measurement, without a physical mass or spacetime interpretation.

DATA AND CODE AVAILABILITY

The extracted terminal radius traces for all 85 seed-condition cases, source-file hashes, configurations, numerical reference calculations, and figure/table rebuild code are available in the versioned repository [10]. The earlier kernel-control summaries and original simulation code are included there. The extract contains the observables needed for the radius analysis, not every original trajectory coordinate or the full historical simulation output.

A single script rebuilds the reported ratios and figures without running new simulations.

Appendix A: Second-Moment and Transport Identities

For normalized point deposition, the new memory is a mixture of the old measure with weight q and a point measure with weight α . The variance identity for this mixture is

$$Q_{n+1} = qQ_n + q|c_n - c_{n+1}|^2 + \alpha|x_{n+1} - c_{n+1}|^2 \quad (\text{A1})$$

$$= qQ_n + q\alpha|x_{n+1} - c_n|^2. \quad (\text{A2})$$

This proves Eq. (10) without a Gaussian assumption. Its stationary expectation and $r_{n+1} = q(x_{n+1} - c_n)$ give Eq. (11). The position-relative variance follows from the innovation independence in Eq. (8). For a centered deposition profile of nonzero variance, the cloud recursion acquires its deposition variance term; the point-deposition result must not be applied unchanged to that model.

For finite history, let h_n be the visible-coordinate response to a unit innovation, with zero prehistory. The centroid response is $b_n = \sum_j w_j h_{n-j}$. The two impulse energies per coordinate are

$$E_r = \sum_{n \geq 0} (h_n - b_n)^2, \quad (\text{A3})$$

$$E_Q = \sum_{n \geq 0} \sum_j w_j (h_{n-j} - b_n)^2. \quad (\text{A4})$$

Multiplying by $d\varepsilon^2$ yields the second moments. The position response itself approaches a nonzero constant, but both centered energies converge. Parseval's identity gives Eqs. (15) and (16); for the latter the weighted sum of squared filter differences simplifies to $1 - |B_H|^2$.

For $H > 1$ and $0 < g < 1$, the noiseless position-history companion map has nonnegative top-row coefficients $(1 - g + gw_0, gw_1, \dots, gw_{H-1})$ summing to one and shift rows below it. The matrix is irreducible and aperiodic. Its eigenvalue one is the translation mode, and all remaining eigenvalues have modulus below one. At $g = 0$, the visible process is a random walk; its finite-window lag differences still have stationary second moments. This

justifies the relative linear reference for the tested gain cells; it is not a stability theorem for the nonlinear Gaussian map.

Expanding the filter near zero frequency gives $B_H(e^{i\omega}) = 1 - i\omega\bar{j}_H + O(\omega^2)$ and $F_H(e^{i\omega}) = i\omega(1 + g\bar{j}_H) + O(\omega^2)$. Thus the large-scale innovation variance is $\varepsilon^2/(1 + g\bar{j}_H)^2$, proving Eq. (17), with

$$\bar{j}_H = \frac{q}{\alpha} - \frac{Hq^H}{1 - q^H}. \quad (\text{A5})$$

The untruncated limit has $\bar{j} = q/\alpha$. The discrete update count remains a model parameter; no physical-time calibration is inferred.

Appendix B: Reproduction and Measurement Limits

The finite reference is evaluated by two implementations: a causal FIFO recurrence with a single unit innovation and a midpoint frequency integral with 131,072 points on $[0, \pi]$. The reference tests include $H = 1$, the zero-gain random-walk covariance obtained from pairwise lag distances, the untruncated limit, and a second (α, H, g) cell distinct from the reported long runs. Their purpose is to test the reference calculation, not to provide new nonlinear simulation evidence.

The terminal block is selected from the stored integer timestamps before any radius ratio is computed. It is the maximal regular suffix, which is 10,001 measurements at unit stride for every reported case. Source-file hashes and the extracted-array archive hash are recorded; a rebuild checks the archive hash. Squared radii are averaged first, then square-rooted; the seed ratios are aggregated last. No instantaneous-radius median is substituted into the RMS comparison.

The two halves of each terminal block are also reported in the machine-readable output. They expose finite-window variability without selecting a favorable subwindow. Since these measurements are correlated, neither the number of stored points nor the preceding N updates supplies an independent sample count. The five-seed IQR is the only uncertainty summary used in the main figures. No statistical equivalence margin or confidence-level claim is attached to the observed discrepancies.

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